

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET E — CASE STUDY & SOURCE-BASED SCENARIOS****CHAPTER 04: EARLY HUMANS AND BEGINNING OF CIVILISATION**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-E-04

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET E - Case Study & Source-Based Scenarios):

1. The question paper comprises **5 In-Depth Case Studies & Source Scenarios** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each source passage, historical inscription, geographic case, and economic scenario carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: ROCK ART & PREHISTORIC ECOLOGY AT BHIMBETKA ROCK SHELTERS

Nestled in the sandstone foothills of the Vindhyan Range in Raisen district, Madhya Pradesh, Bhimbetka contains more than 750 rock shelters, over 500 of which are adorned with prehistoric paintings. Discovered by archaeologist V. S. Wakankar in 1957, Bhimbetka spans a continuous cultural sequence from the Lower Palaeolithic through the Mesolithic to the historic period. The most vibrant art belongs to the Mesolithic phase. Prehistoric hunter-gatherers manufactured natural mineral pigments: red haematite (iron oxide), white kaolin (china clay), green copper minerals, and charcoal, bound together with animal fat and tree sap. The paintings depict dynamic scenes of collective hunting expeditions: hunters wielding microlith-tipped spears and bows, communal dances with dancers holding hands in rhythmic rows, women gathering tubers and honeycombs, and diverse wild fauna including bison, tigers, rhinoceroses, and deer. Notably, no agricultural crops, domestic cattle ploughs, or metal swords appear in the early hunter-gatherer strata, capturing a vivid empirical snapshot of humanity's pre-agrarian existence.

Q.1 Who discovered the world-famous prehistoric rock shelters of Bhimbetka in 1957?

- (A) Sir Mortimer Wheeler (B) Dr. V. S. Wakankar
(C) Alexander Cunningham (D) James Prinsep

Q.2 What natural mineral substance was ground by prehistoric artists to obtain the rich red pigment seen on the cave walls?

- (A) Crushed lapis lazuli gemstones (B) Haematite (iron oxide mineral, locally known as geru)
(C) Modern chemical acrylic oil paints (D) Powdered gold nuggets

Q.3 What subsistence activities are predominantly depicted in the Mesolithic rock paintings of Bhimbetka?

- (A) Industrial factory textile weaving and tractor ploughing (B) Collective hunting of wild game, honey gathering, and communal dancing
(C) Coin minting and tax collection by royal imperial officers (D) Construction of multi-storey brick skyscrapers

Q.4 The absence of plough agriculture, draft bullocks, and metal swords in the early Bhimbetka paintings proves that:

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| (A) The artists belonged to a pre-agrarian, Stone Age hunter-gatherer society | (B) The artists were modern tourists playing a prank |
| (C) The artists were blind and could not see domesticated animals | (D) The cave paintings were created in the 20th century |

Q.5 What diagnostic tool technology emerged during the Mesolithic period, utilized by hunters shown in the paintings?

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| (A) Heavy unpolished quartzite hand-axes | (B) Microliths—tiny, precisely knapped stone blades hafted onto wood or bone handles |
| (C) Iron swords and steel muskets | (D) Bronze wheels and cannons |

CASE STUDY 2: CIVIL ENGINEERING & WATER MANAGEMENT ARCHITECTURE AT DHOLAVIRA

Dholavira, situated on Khadir Bet island in the Great Rann of Kutch, Gujarat, represents one of the five largest Harappan metropolises. Unlike other mature Harappan settlements constructed of baked mud-bricks, Dholavira's monumental architecture was constructed using dressed sandstone blocks. The city was divided into three distinct fortified sectors: the Citadel (Castle and Bailey), the Middle Town, and the Lower Town. Located in a hyper-arid saline environment with erratic annual rainfall of under 300 mm, Dholavira engineers designed an extraordinary hydraulic engineering marvel. The settlement was bounded by two seasonal ephemeral streams: the Manhar on the south and the Mansar on the north. Engineers constructed massive stone check-dams across these torrents, diverting seasonal stormwater through masonry canals into a cascading circuit of 16 gigantic rock-cut reservoirs encircling the city. The largest reservoir measured 73 metres long, 29 metres wide, and over 10 metres deep, equipped with broad stone staircases. This ingenious system stored nearly 300,000 cubic metres of fresh rainwater, sustaining an estimated urban population of 20,000 through severe multi-year droughts.

Q.6 Unlike Mohenjo-daro and Harappa, what distinctive building material was predominantly used in Dholavira's monumental structures?

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| (A) Burnt red clay bricks only | (B) Dressed limestone and sandstone blocks with mud-mortar |
| (C) Hollow bamboo stems and palm leaves | (D) Solid iron and steel girders |

Q.7 What unique tripartite (three-tier) urban layout distinguishes Dholavira from typical Harappan dual-citadel cities?

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| (A) A single undivided village street | (B) A three-tier division into Citadel, Middle Town, and Lower Town |
| (C) Ten concentric circular wooden palisades | (D) A grid of floating houseboats on the ocean |

Q.8 How did Dholavira engineers harvest and conserve freshwater in the arid Rann of Kutch?

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| (A) By constructing check-dams across seasonal streams and channeling water into massive rock-cut reservoirs | (B) By importing fresh water daily in ceramic jars from Mesopotamia |
| (C) By digging an underground tunnel to the distant Ganga River | (D) By boiling saline ocean water over charcoal fires |

Q.9 What major environmental challenge did Dholavira's sophisticated hydraulic system overcome?

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| (A) Continuous polar freezing of all surface streams | (B) Arid desert conditions with low, erratic monsoon rainfall and prolonged droughts |
| (C) Frequent eruptions of active lava volcanoes | (D) Daily tidal tsunamis from the Arctic Ocean |

Q.10 What famous epigraphic discovery was uncovered at the northern gateway of Dholavira's Citadel?

- (A) A trilingual inscription written in Greek, Latin, and Sanskrit (B) A large 'signboard' consisting of 10 large Harappan symbols inlaid with white crystalline gypsum
- (C) A royal biography of Emperor Ashoka carved on a granite pillar (D) A collection of paper manuscript scrolls detailing city taxes

CASE STUDY 3: SANITARY ENGINEERING & DRAINAGE MECHANICS OF MOHENJO-DARO

The urban layout of Mohenjo-daro in Sindh represents the pinnacle of public sanitary engineering in the Bronze Age ancient world. Excavations directed by Ernest Mackay and John Marshall demonstrated that civic planners laid out the city on an orthogonal gridiron plan, with wide avenues running strictly north-south and east-west, intersecting at 90-degree right angles. Every residential home, whether small two-room quarters or spacious multi-room villas with central courtyards, possessed a paved brick bathing platform. Waste water from bathrooms and latrines flowed through terracotta clay pipes embedded in house walls into small household drainage channels. These fed directly into covered brick drains running beneath street pavements. The street drains were constructed of precision-moulded baked bricks laid with gypsum and lime mortar. They were covered with dressed limestone slabs or loose baked bricks that could be lifted by municipal sanitation workers for periodic cleaning. Sump pits and settling traps were installed at regular intervals along street channels to collect solid sediment before clear water drained out of the city gates.

Q.11 What geometric pattern guided the architectural street planning of Mature Harappan cities like Mohenjo-daro?

- (A) Irregular meandering alleys winding without any design (B) An orthogonal gridiron plan where main avenues intersected at right angles (90°)
- (C) Concentric spider-web circles with a palace at the center (D) A single linear road hugging the riverbank

Q.12 How were household waste waters conveyed into the public street sewers in Mohenjo-daro?

- (A) They were dumped openly from high balconies into the middle of the street (B) Through covered terracotta drain pipes running inside exterior walls into street sewer channels
- (C) They were stored in bedroom pots and never discarded (D) Household drains did not exist anywhere in the ancient city

Q.13 Why were Harappan street drains equipped with loose stone covers and regular sump pits?

- (A) To hide gold and silver coins from foreign invaders (B) To facilitate municipal maintenance, inspections, and cleaning of accumulated solid silt
- (C) To catch wild rats for household food consumption (D) To allow rainwater to leak out and flood residential rooms

Q.14 What mortar material was utilized to ensure waterproofing and strength in Harappan sanitary brickwork?

- (A) Soft desert sand mixed with animal blood (B) Gypsum, lime, and mud mortar
- (C) Liquid petroleum and modern Portland cement (D) Raw clay without baking

Q.15 What does the sophisticated, uniform drainage system found across all Harappan cities signify to historians?

- (A) A complete lack of civic organization or hygiene (B) A high degree of centralized municipal governance, civic hygiene, and urban standardization
- (C) That each household built drains randomly without any municipal rules (D) That Harappan cities were exclusively used as religious monasteries

CASE STUDY 4: INTERNATIONAL MARITIME TRADE & PORT ENGINEERING AT LOTHAL

Excavations conducted by S. R. Rao at Lothal, situated along the Bhogavo River in the Gulf of Khambhat, Gujarat, unearthed unambiguous archaeological evidence of a thriving Harappan maritime trading port. The crowning discovery at Lothal is a massive trapezoidal basin measuring 217 metres in length, 36 metres in width, and over 4 metres deep, constructed entirely of kiln-burnt bricks. Hydraulic analysis confirms this basin was an engineered tidal dockyard. At high tide, seawater entered the dockyard through an inlet channel connected to the river estuary. An adjustable wooden lock-gate operated in a vertical brick groove, allowing ships to enter and trapping water at low tide so vessels remained afloat while loading and unloading cargo. Adjoining the dockyard was a massive raised brick warehouse (consisting of 64 cubic brick blocks) where imported raw materials and export goods were stored. Archaeologists recovered over 200 clay sealings bearing impressions of Harappan seals with woven cloth markings on their reverse, alongside a distinctive circular steatite 'Persian Gulf' seal, proving direct maritime commercial ties with Dilmun (Bahrain), Magan (Oman), and Meluhha (the Indus valley).

Q.16 What monumental engineering structure was uncovered at Lothal by archaeologist S. R. Rao?

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| (A) A Roman amphitheater for chariot races | (B) A kiln-burnt brick tidal dockyard with an inlet channel and lock-gate mechanism |
| (C) A stone pyramid housing royal mummies | (D) An astronomical glass telescope observatory |

Q.17 How did the tidal dockyard maintain sufficient water depth to keep cargo ships afloat during low tide?

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| (A) Workers poured water into the basin with leather buckets | (B) A vertical sluice lock-gate trapped high-tide seawater inside the brick basin |
| (C) The dockyard was built on top of a subterranean freshwater geyser | (D) Ships were hoisted out of the water using giant steam cranes |

Q.18 What archaeological evidence found in the Lothal warehouse proves that goods were packaged and sealed for commercial transport?

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| (A) Clay sealings bearing Harappan seal motifs on the front and woven cloth/cord impressions on the reverse | (B) Paper cardboard boxes with printed barcodes |
| (C) Iron padlocks inscribed with Roman letters | (D) Plastic shipping containers labeled in English |

Q.19 The discovery of a distinctive circular 'Persian Gulf' button seal at Lothal indicates direct maritime contact with:

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| (A) Ancient Mesoamerica | (B) Dilmun (modern Bahrain) and the Persian Gulf trade network |
| (C) The Scandinavian Viking kingdoms | (D) The Australian continent |

Q.20 In ancient Mesopotamian cuneiform trade records, which foreign land is identified as the Harappan/Indus region?

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| (A) Meluhha | (B) Atlantis |
| (C) Magan | (D) Carthage |

CASE STUDY 5: THE DE-URBANISATION & DECLINE OF THE MATURE HARAPPAN CIVILISATION

Around 1900 BCE, the mature urban phase of the Harappan Civilisation entered a pronounced period of decline and transformation, often termed the Late Harappan or Post-Urban phase. For decades, early colonial historians like Mortimer Wheeler postulated a violent cataclysm, attributing the collapse to 'Aryan invasions' based on an unburied cluster of

skeletons found in Mohenjo-daro. However, modern forensic anthropology, stratigraphic re-examination, and palaeo-climatic sediment cores have completely dismantled the invasion myth. Skeletal remains show no traumatic weapon blade wounds, and the bodies belong to different stratigraphic periods. Instead, multi-proxy environmental data reveals that the 4.2k BP climatic aridification event caused a severe, prolonged weakening of the South Asian summer monsoon. Concurrently, tectonic shifts altered river courses: the mighty Ghaggar-Hakra (often identified with the Vedic Sarasvati) dried up as its glacial headwaters were captured by the Yamuna and Sutlej. Faced with failing agricultural surpluses, urbanites abandoned the great brick cities of the Indus valley and migrated eastward and southward into Gujarat, the upper Gangetic plain, and western Uttar Pradesh, establishing smaller rural agrarian settlements.

Q.21 Why has modern archaeological science rejected Mortimer Wheeler's 'Aryan invasion' theory for the end of Harappa?

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| <p>(A) Ancient written diaries were found saying Wheeler was wrong</p> <p>(C) All Harappan cities were destroyed by modern nuclear bombs</p> | <p>(B) Forensic analysis shows skeletons show no battle weapon trauma, belong to different stratigraphic layers, and show zero sign of military conquest</p> <p>(D) Harappan civilization never existed in the first place</p> |
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Q.22 What major environmental and hydrographic crisis is confirmed by palaeo-climatic studies around 1900 BCE?

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| <p>(A) A permanent rise of global oceans by 500 meters</p> <p>(C) A sudden advance of polar ice glaciers covering all of Gujarat</p> | <p>(B) A prolonged multi-century climatic drought coupled with the drying up of the Ghaggar-Hakra river system</p> <p>(D) A massive asteroid impact that vaporized the Arabian Sea</p> |
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Q.23 What characterized the Late Harappan (Post-Urban) cultural phase compared to the Mature Harappan phase?

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| <p>(A) Expansion of larger, wealthier cities with skyscrapers and subways</p> <p>(C) Establishment of an absolute world empire conquering Egypt and China</p> | <p>(B) De-urbanization: abandonment of large cities, disappearance of the standardized script and chert weights, and a shift to rural farming villages</p> <p>(D) Complete adoption of iron technology and modern English schools</p> |
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Q.24 Where did Harappan populations migrate as the central Indus and Sarasvati river valleys dried up?

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| <p>(A) They emigrated exclusively to the North Pole</p> <p>(C) They boarded wooden sailing ships and moved to South America</p> | <p>(B) They migrated eastward and southward into Gujarat, Punjab, Haryana, and the upper Gangetic plain</p> <p>(D) They moved into deep subterranean gold mines</p> |
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Q.25 What fundamental lesson does the rise and transformation of the Indus Civilisation offer to contemporary societies?

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| <p>(A) That advanced human cities are completely immune to environmental or climatic changes</p> <p>(C) That municipal hygiene and drainage are totally useless in human settlements</p> | <p>(B) That complex urban civilisations depend heavily on sustainable ecological balances, stable water systems, and agricultural security</p> <p>(D) That building cities near rivers always causes instant human extinction</p> |
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CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & CASE STUDY ANALYTICAL EXPLANATIONS (SET E)

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.1	(B)	Archaeologist Dr. Vishnu Shridhar Wakankar discovered the Bhimbetka rock shelters while surveying the Vindhyan hills in 1957.
Q.2	(B)	Red pigments in prehistoric cave art were derived from haematite (red ochre / iron oxide) mixed with water, sap, and animal fat.
Q.3	(B)	Mesolithic rock art reflects the life of nomadic foragers: hunting large game with spears/bows, foraging wild honey, and performing communal rituals.
Q.4	(A)	The material culture illustrated matches the technological and economic baseline of prehistoric Stone Age hunting and gathering societies.
Q.5	(B)	Mesolithic culture is defined by microlithic tools—minute geometric blades mounted on shafts to make composite arrows, sickles, and spears.
Q.6	(B)	Dholavira is renowned for using quarried, finely dressed stone masonry alongside mud-bricks for its massive fortification walls and reservoirs.
Q.7	(B)	Dholavira exhibits a unique tripartite urban plan consisting of a monumental Citadel, an intermediate Middle Town, and an expansive Lower Town.
Q.8	(A)	Dholavira's engineers built masonry dams across the Manhar and Mansar torrents to divert storm runoff into an interconnected network of 16 huge reservoirs.
Q.9	(B)	In the arid saline desert of Kutch, large-scale rainwater storage was essential for urban survival during recurrent droughts.
Q.10	(B)	The Dholavira Signboard consists of 10 large, beautifully formed Harappan characters made of inlaid gypsum crystal, set into a wooden board.
Q.11	(B)	Harappan urban settlements were planned on an orthogonal gridiron system, with wide, straight streets intersecting at precise right angles.
Q.12	(B)	Every house had dedicated paved bathing platforms with pipes discharging into covered municipal drains along the street.
Q.13	(B)	Removable stone slab covers and inspection sump pits allowed civic sanitation workers to clean out silt and prevent municipal sewer clogging.
Q.14	(B)	Harappans used finely prepared gypsum and lime mortar to waterproof brick drains and structures like the Great Bath.
Q.15	(B)	Uniform covered drainage across distant cities proves the presence of highly organized municipal authorities focused on public sanitation.
Q.16	(B)	Lothal contains the world's earliest known tidal dockyard, engineered with burnt-brick walls, inlet sluices, and water-retaining lock gates.
Q.17	(B)	A wooden sluice gate inserted into vertical masonry grooves trapped high-tide water inside the basin, preventing ships from grounding during low tide.
Q.18	(A)	Clay sealings stamped over rope knots and burlap cloth on trade packages verify commercial sealing and origin authenticity.
Q.19	(B)	Circular Persian Gulf button seals link Lothal's port directly with commercial transit ports in Bahrain and the Gulf region.

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.20	(A)	Mesopotamian Akkadian and Sumerian texts refer to the Indus Civilisation as 'Meluhha', a land of carnelian, lapis lazuli, copper, and ivory.
Q.21	(B)	Forensic pathology of skeletons at Mohenjo-daro found no blade trauma; the burials were haphazard over time, disproving a single massacre.
Q.22	(B)	Climatic drying (the global 4.2k BP arid event) and tectonic shifts that diverted headwaters away from the Ghaggar-Hakra caused urban collapse.
Q.23	(B)	The Late Harappan phase saw de-urbanization: standard weights, writing, and baked-brick architecture disappeared as populations decentralized into villages.
Q.24	(B)	Archaeological surveys show a proliferation of small Late Harappan rural agricultural sites in Gujarat, Haryana, and the Yamuna-Ganga doab.
Q.25	(B)	The Harappan experience demonstrates the critical dependency of complex urban civilizations on environmental sustainability and water security.